SG No. 187 D_{3h}^{1} $P\bar{6}m2$ [hexagonal]

* plus set: +[0, 0, 0]

Table 1: Wyckoff site: `1a`, site symmetry: `-6m2`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `1b`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `1c`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 4: Wyckoff site: `1d`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 5: Wyckoff site: `1e`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 6: Wyckoff site: `1f`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 7: Wyckoff site: `2g`, site symmetry: `3m.`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,7,8,9]`
2	$[0, 0, -z]$	`[4,5,6,10,11,12]`

Table 8: Wyckoff site: 2h, site symmetry: 3m.

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1,2,3,7,8,9]
2	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	[4,5,6,10,11,12]

Table 9: Wyckoff site: 2i, site symmetry: 3m.

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	[1,2,3,7,8,9]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[4,5,6,10,11,12]

Table 10: Wyckoff site: 3j, site symmetry: mm2

No.	position	mapping
1	$[x,\ -x,\ 0]$	[1,4,7,10]
2	$[x,\ 2x,\ 0]$	[2,5,9,12]
3	$[-2x,\ -x,\ 0]$	[3,6,8,11]

Table 11: Wyckoff site: 3k, site symmetry: mm2

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{2}\right]$	[1,4,7,10]
2	$\left[x,\ 2x,\ \frac{1}{2}\right]$	[2,5,9,12]
3	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	[3,6,8,11]

Table 12: Wyckoff site: 6l, site symmetry: m..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,4]
2	$[-y,\ x-y,\ 0]$	[2,5]
3	$[-x+y,\ -x,\ 0]$	[3,6]
4	$[-y,\ -x,\ 0]$	[7,10]
5	$[-x+y,\ y,\ 0]$	[8,11]
6	$[x,\ x-y,\ 0]$	[9,12]

Table 13: Wyckoff site: 6m, site symmetry: m..

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,4]
2	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[2,5]
3	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[3,6]
4	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[7,10]
5	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	[8,11]

continued ...

Table 13

No.	position	mapping
6	$\left[x,\, x-y,\, \frac{1}{2}\right]$	`[9,12]`

Table 14: Wyckoff site: `6n`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\, -x,\, z]$	`[1,7]`
2	$[x,\, 2x,\, z]$	`[2,9]`
3	$[-2x,\, -x,\, z]$	`[3,8]`
4	$[x,\, -x,\, -z]$	`[4,10]`
5	$[x,\, 2x,\, -z]$	`[5,12]`
6	$[-2x,\, -x,\, -z]$	`[6,11]`

Table 15: Wyckoff site: `12o`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x-y,\, z]$	`[2]`
3	$[-x+y,\, -x,\, z]$	`[3]`
4	$[x,\, y,\, -z]$	`[4]`
5	$[-y,\, x-y,\, -z]$	`[5]`
6	$[-x+y,\, -x,\, -z]$	`[6]`
7	$[-y,\, -x,\, z]$	`[7]`
8	$[-x+y,\, y,\, z]$	`[8]`
9	$[x,\, x-y,\, z]$	`[9]`
10	$[-y,\, -x,\, -z]$	`[10]`
11	$[-x+y,\, y,\, -z]$	`[11]`
12	$[x,\, x-y,\, -z]$	`[12]`